GADS: Subsystem Definition & Interaction Specification
Version 2.1

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Core Philosophy: Emergence through Coupling

Subsystems in a GADS operate on distinct timescales; the system's global dynamics emerge from their coupling rather than any single component. The system is a feedback loop where fast phenotypic exploration is constrained and directed by slow representational evolution.

This document presents the theory in three layers. First, the axiomatic derivation establishes why each subsystem must exist by tracing a chain from structural axiom to failure mode to required subsystem. Second, the subsystem specification describes what each subsystem is and how it operates. Third, the dynamics section describes the emergent behavior that arises from subsystem coupling.

1 Axiomatic Derivation of Subsystem Architecture

The following eight derivation chains establish that a GADS necessarily consists of exactly six subsystems. Each chain begins with a structural axiom, derives a constraint via lemma, and concludes with the subsystem that the constraint forces into existence. Removing any subsystem produces a specific, identified failure mode. No additional subsystem is required; every failure mode introduced by the axioms is addressed by one of the six.

1.1 Chain 1: Structural Locality

Axiom: All operations have local scope. No operation may modify objects outside its own system boundary.

Lemma: No global controller or semantic oracle can exist. Nothing has the authority or reach to enforce coherence top-down.

Therefore: All coordination must occur through explicit inter-subsystem interfaces. Coherence is a product of coupled local interactions, not central planning.

1.2 Chain 2: Persistent Semantic Store

Axiom: Compositions of symbols can be assigned identity tokens that are reusable, losslessly recoverable, and deduplicated.

Lemma: A composition's identity must remain decodable for as long as it exists in the system. The semantic store must maintain a valid mapping from token to content at all times that the token is referenced, whether by organisms in the population or by other compositions. The store does not guarantee persistence; it guarantees that anything still referenced can still be decoded.

Therefore: A persistent semantic store must exist (MG). It maintains the mapping from identity tokens to their compositions and supports recursive expansion (decode).

1.3 Preliminary Definitions: $\mathcal{S}(t)$ and $\mathcal{A}(t)$

Two concepts must be distinguished that prior versions of the theory conflated. Chain 3 depends structurally on the relationship between them, so they are defined here before the derivation that uses them.

Service set $\mathcal{S}(t)$: The set of all compositions currently referenced by at least one organism in the population at time t. It has no fixed capacity. It is not a container—it is a description of what is in service. A composition is in the service set if and only if at least one organism in P_t contains it as a top-level identifier token. The service set is not directly observable by any subsystem; its contents are revealed only through decode requests made by the population for top-level tokens during fitness evaluation. Each such decode request emits exactly one participation signal for the referenced identifier. Identifiers encountered during recursive expansion of a decoded composition do not independently generate participation signals.

Register container $\mathcal{A}(t)$: The bounded-capacity container specified by the PFOL as the Active set. It holds at most π_{14} compositions ordered by last participation event. Residency in $\mathcal{A}(t)$ provides a buffer against displacement into the deletion basket. A composition may be in $\mathcal{A}(t)$ without being in $\mathcal{S}(t)$ (it occupies a register slot but no organism references it), and a composition may be in $\mathcal{S}(t)$ without being in $\mathcal{A}(t)$ (it is referenced by organisms but has been displaced to the deletion basket).

$\mathcal{S}(t)/\mathcal{A}(t)$ Configuration Space

The four possible configurations of a composition with respect to $\mathcal{S}(t)$ and $\mathcal{A}(t)$ are:

(a) **In both $\mathcal{S}(t)$ and $\mathcal{A}(t)$:** In service and buffered. The composition is referenced by at least one organism and holds a register slot. It will receive participation signals during decode and refresh its register position.

(b) **In $\mathcal{S}(t)$ but not $\mathcal{A}(t)$:** In service but displaced. The composition is referenced by organisms but has been moved to the deletion basket. The next decode event involving a surviving organism that carries it will generate a participation signal, rescuing it back to the register.

(c) **In $\mathcal{A}(t)$ but not $\mathcal{S}(t)$:** Occupying a register slot but not referenced by any organism. This is a False-signal composition. It receives no participation signals and drifts toward the displacement boundary.

(d) **In neither $\mathcal{S}(t)$ nor $\mathcal{A}(t)$:** The composition has been deleted and its identifier returned to the reuse pool, or it is currently aging in the deletion basket with no mechanism to generate a rescue signal.

These configurations are exhaustive. Chain 3, Section 4.3, and Section 4.4 depend on this vocabulary.

Note: The PFOL symbol $\mathcal{A}(t)$ continues to refer to the register container. The service set $\mathcal{S}(t)$ is a theoretical state required by the GADS theory for reasoning about subsystem interaction under different population conditions. The PFOL does not formalize $\mathcal{S}(t)$ because the canonical implementation realizes its effects through the participation signal during decode, not because $\mathcal{S}(t)$ is secondary or optional. Both $\mathcal{A}(t)$ and $\mathcal{S}(t)$ are needed: $\mathcal{A}(t)$ is the mechanism; $\mathcal{S}(t)$ is the state that the mechanism acts in response to.

1.4 Chain 3: Finite Active Capacity and the Participation Register

Axiom: The system has bounded memory for active compositions.

Lemma: The number of compositions will eventually exceed capacity because creation is ongoing and there is no intrinsic upper bound on captures.

Lemma: The system must therefore enforce a boundary between compositions that are retained and compositions that are displaced. This requires a mechanism that determines displacement order.

Valuation Constraint

Lemma (from Chain 8): No subsystem can independently evaluate the worth of a composition. The only available signal is whether a composition was decoded during population evaluation—a binary fact. The composition was either present in a decoded organism or it was not. Frequency of use within a generation, fitness correlation, hierarchical depth, or any other metric would require interpretive machinery that violates structural locality (Chain 1) and the population's exclusive role as the valuation channel (Chain 8).

Lemma (Binary valuation event): A participation signal is not an assignment of scalar value. It is proof of existence within the population's realized decoding history for that generation. The system detects only the presence or absence of participation—not magnitude, not ranking, not degree.

Persistence Requirement

Lemma (Persistence mechanism necessity): Under finite capacity, absence of participation must have structural consequences once new compositions enter. However, because the signal is binary and non-graded, the system cannot rank compositions by value.

Therefore, the mechanism that governs persistence must: not count participation frequency, not assign weights, not compute scores, not interpret magnitude. It must only preserve the temporal record of participation events.

Asymmetric Update Derivation

Lemma (Asymmetric update rule): Since the participation signal is the sole interface through which environmental pressure reaches the representational layer (Chain 8), it is the only event that can legitimately modify displacement ordering. The only admissible positional update is: on participation event, move to the retention boundary. No other event modifies position.

This produces an unavoidable asymmetry between two classes of composition:

True-signal class: Compositions that receive at least one participation event in a generation. All such compositions are moved to the retention boundary. A composition decoded once and a composition decoded fifty times end up at the same position. Ordering among True-signal compositions is neutral—no composition in this class is at greater risk of displacement than any other, regardless of decode frequency.

False-signal class: Compositions that receive zero participation events. These compositions do not change position. They remain wherever their last participation event placed them. Their relative ordering among other False-signal compositions is determined entirely by when they last participated. Ordering among False-signal compositions is the sole determinant of their proximity to the displacement boundary.

Therefore: Recency ordering is *selectively neutral*. It is neutral within the True-signal class (no valuation distinction among participating compositions). It is operative within the False-signal class (determining displacement priority among non-participating compositions). The ordering does not assign value. It makes absence-of-participation structurally consequential.

Service Set and Register Interaction

The register container $\mathcal{A}(t)$ and the service set $\mathcal{S}(t)$ are distinct (Section 1.3). Their relationship determines the character of the metabolic dynamics. Three cases arise:

Case 1: Full cycling under sustained congestion. $|\mathcal{S}(t)| \geq \pi_{14}$. Every composition in the register is in the service set. All register residents receive participation signals each generation. The True-signal class spans the entire register. Ordering is irrelevant—there are no False-signal compositions in the register to sort. Displacement from new captures sends compositions to the deletion basket, but because they are in $\mathcal{S}(t)$, they generate rescue signals and return to the register.

Case 2: Sustained congestion with genuine disuse. Some compositions fall out of $\mathcal{S}(t)$—no organism references them. Congestion is active, so new captures displace them immediately (they are in the False-signal class and therefore closest to the displacement boundary). Once in the deletion basket, they cannot generate rescue signals because they are not in $\mathcal{S}(t)$. They age to the threshold and are deleted. Ordering accelerates their displacement but does not determine their fate—their absence from $\mathcal{S}(t)$ does.

Case 3: Partial cycling. $|\mathcal{S}(t)| < \pi_{14}$. The population is not using enough compositions to fill the register. Some register slots hold compositions not in $\mathcal{S}(t)$. Congestion may be intermittent or absent. During lulls, no displacement occurs, and these unused compositions sit in the register undisturbed. However, because they receive no participation signals, they do not refresh. Compositions in $\mathcal{S}(t)$ continue to refresh and accumulate at the retention boundary. The False-signal compositions drift toward the displacement boundary—not through any active process, but because refresh events by True-signal compositions shift the relative ordering.

When congestion resumes, the False-signal compositions are structurally guaranteed to be at the displacement boundary. They are displaced first. The ordering preserved non-participation history across the lull, ensuring that when the metabolic pipeline reactivates, it does not start from zero information about which compositions have been unused.

This is where recency ordering earns its necessity. Without persistent ordering, displacement during resumed congestion would be arbitrary among all compositions not decoded in the most recent generation. The accumulated evidence of sustained non-participation would be lost. The ordering is not a judge—it is a lens that makes the population's binary signal legible to the displacement mechanism across temporal gaps in congestion.

1.5 Chain 4: Grace Period Under Capacity-Driven Eviction

Axiom: Eviction from the working set is driven by capacity pressure, not by evidence of absence from the population.

Lemma: An evicted abstraction may still be present in organisms that have not been decoded since eviction. Immediate deletion would destroy an abstraction the population is actively using.

Lemma: Proof of absence requires one full independent evaluation cycle in which every organism is decoded and the abstraction receives zero usage signals. This takes a minimum of 2 generation ticks. Age 0 means just evicted. Age 1 is ambiguous because the evaluation cycle that triggered eviction may overlap. Age 2 means one full cycle with confirmed zero use.

Therefore: A temporal buffer must exist between eviction and deletion (DB) with a minimum age threshold of 2.

1.6 Chain 5: Information-Preserving Deletion

Axiom: Structural rewrites must not alter phenotypic expression. The decoded output of every organism and every composition must be identical before and after deletion.

Lemma: A deleted composition may be referenced by other compositions. Removing it without replacement would corrupt their decoded output.

Therefore: Deletion must inline, replacing all references to the deleted composition within other compositions with its expanded content, before freeing the identifier. This is the only local operation that preserves phenotypic identity without modifying the population, which would violate locality (Chain 1).

1.7 Chain 6: Identifier Recycling

Axiom: The identifier space is finite or practically bounded. New compositions require identifiers. Deleted compositions free identifiers.

Lemma: Without recycling, the system eventually exhausts its identifier space and can no longer create new compositions. Representational growth halts permanently.

Therefore: An allocation mechanism must exist (EA) that recycles freed identifiers while maintaining referential integrity. No identifier is reused while any reference to it exists, which is guaranteed by inlining in Chain 5.

1.8 Chain 7: Recency-Biased Selection

Axiom: New compositions are continually created from the current population's state via capture. The population's location in the search space changes over time.

Lemma: Older compositions were formed under fitness conditions that may no longer hold. As the library grows, uniform sampling increasingly selects abstractions that are irrelevant to the current population.

Lemma: This wastes mutation events on outdated material and pollutes the population with structures that selection has already moved past.

Therefore: Selection of compositions for use in mutations must preferentially favor newer compositions. A chronological record of creation order must exist (MCO), and the selection distribution must weight recent entries more heavily.

1.9 Chain 8: Population as Energetic Engine

Axiom: A selection criterion exists that differentially evaluates configurations. Differential survival is the only channel through which the system determines the value of an abstraction.

Lemma: No subsystem can independently evaluate whether an abstraction is useful. The MG does not know fitness. The Participation Register does not know fitness. The DB does not know fitness. Only the population, through surviving organisms being decoded during fitness evaluation, generates usage signals.

Lemma (Interface-Scoped Participation): A participation signal may be emitted only when the population explicitly requests the decode of a composition identifier that appears as a top-level token in a surviving organism. Each such decode request emits exactly one Usage(id) event for the referenced identifier. Recursive expansion performed internally by the MG during Decode(id)

does not emit additional usage signals. Identifiers encountered during recursive expansion are not independently credited with participation.

Lemma: An abstraction persists if and only if organisms explicitly carrying its identifier as a top-level token survive selection and are decoded. Containment within another composition does not constitute independent participation. This is the sole feedback mechanism connecting representational identity to environmental pressure.

Therefore: The population (POP) is the energetic engine. It emits two signals at the subsystem interface:

- **Capture(mapping):** Triggered by successful motif discovery in the population.
- **Usage(id):** Triggered once per top-level identifier token decoded during fitness evaluation of surviving organisms.

These signals are the only inputs the metabolic subsystems (Participation Register, DB, EA) and the slow subsystems (MG, MCO) receive. No subsystem receives information about internal structural decomposition beyond what is exposed by these interface-level signals.

1.10 Closure Argument

These eight chains produce six subsystems: MG, Participation Register, DB, EA, MCO, POP. Each subsystem exists because a specific axiom forces it. No subsystem is optional; removing any one produces a specific, identified failure mode. No additional subsystem is required; every failure mode introduced by the axioms is addressed by one of the six.

The interaction pipelines (metabolic and generative) are the interfaces required by Chain 1 (locality). The timescale separation (fast, metabolic, slow) is a consequence of the different rates at which POP operates (per mutation), Participation Register/DB/EA operate (per generation), and MG/MCO change (per capture or death).

Under the stated axioms, the system is closed under the operators defined by those axioms. Every resource (identifiers, capacity slots, boundary markers) is finite and recycled within the system. Every signal (usage, capture) originates internally from population decode and mutation. Every structural change (creation, eviction, deletion, inlining) preserves phenotypic identity. This closure is relative to the current axiom set and does not preclude the discovery of additional subsystems forced by axioms not yet identified, or further decomposition of existing axioms into finer structure.

This is the theory of GADS. Everything that follows, including phase regimes, dynamics, and specific parameter settings, is consequence, not definition.

2 The Boundary Mechanism

Before describing the subsystems in detail, one structural mechanism requires explicit clarification: the boundary mechanism (delimiters). Boundaries are not a subsystem. They are a structural feature of the population encoding that the subsystems depend on in order to preserve locality during abstraction.

Structural Role: The population (POP) consists of structured sequences on which local operators act. The boundary mechanism introduces paired structural markers that partition each sequence into disjoint regions. These regions define distinct operator-admissibility phases.

Operators may be phase-dependent. In particular:

- Certain structural transformations are admissible only within boundary-delimited regions.
- Certain structural transformations are restricted from acting across boundary interiors.

Boundaries therefore induce operator asymmetry without introducing any global controller or interpretive mechanism. They define where specific operations may legally occur while preserving the structural locality constraint (Chain 1).

Boundaries reside entirely within the population layer. They do not persist independently of organisms and are not stored in the semantic store (MG). They are features of the fast-timescale medium only.

Atomicity and Integrity: A boundary consists of a paired opening and closing marker. These markers are structurally atomic under all operators: they are created and removed only as paired units and cannot be independently altered. This guarantees well-formed boundary structure under mutation, recombination, capture, and open operations without requiring any global validation mechanism.

Boundary Creation and Removal: Boundary markers are dynamically introduced and eliminated by admissible local operations within the population.

- Boundary creation: Certain mutation operations may introduce a new paired boundary into an unbounded region.
- Boundary removal: Certain operations may eliminate an existing boundary pair, either directly or as a consequence of abstraction.

Boundary markers are always created and removed as paired units. No operator may create or destroy a single boundary marker independently. As a result, the total boundary mass of the population evolves over time.

Boundary Mobility: Boundary markers are not fixed in position. Under admissible local operators in the population layer, boundary markers may change position relative to neighboring symbols. This mobility allows the extent of a delimited region to expand or contract over time, incorporating adjacent material or releasing previously enclosed material.

Mobility arises from local operator application alone. It does not introduce semantic interpretation, valuation, or global coordination. It is a geometric property of the population medium.

Non-Interpenetration Constraint: Boundary pairs may not cross, interpenetrate, or pass through one another. Two boundaries may approach and become adjacent, but their relative ordering along the sequence is preserved. This guarantees that delimited regions remain disjoint and prevents structural inversion analogous to two cell walls occupying the same space.

No-Nesting Constraint: Boundary markers do not nest within other boundary markers.

Allowing nesting would require additional admissibility ordering rules (e.g., inner versus outer capture priority), increasing structural complexity without expanding representational capability. Hierarchical structure is already provided by compositional identity (Chain 2). Recursion occurs through metagene expansion, not through boundary stacking.

Excluding nesting preserves minimal structural geometry while retaining full expressive power.

Relationship to Capture: Capture is admissible only on boundary-delimited regions satisfying minimal structural extent. Capture never selects arbitrary substrings of an organism. It acts only on material that has already been structurally delimited.

When capture occurs, the bounded region is replaced by an identity token and the boundary markers are removed. A composition therefore represents nothing more than a previously realized bounded region of population material. It encodes no intrinsic semantic interpretation and carries

no independent valuation. Boundaries define the locus of abstraction while preserving structural locality.

Relationship to Open: The open operator is context-sensitive.

- Outside a boundary, open introduces a new boundary pair around the expanded content.
- Inside a boundary, open inserts expanded content without adding additional markers, preserving the no-nesting constraint.

This asymmetry ensures that recursive expansion remains well-formed without requiring nested boundary structures.

Dynamical Implication: Because capture consumes boundary-delimited regions and boundary-creation operations regenerate them, and because boundaries are mobile under local operators, boundary density and boundary geometry within the population are dynamically regulated. Mobility allows boundaries to negotiate local corridor space, expand to incorporate adjacent material, or contract under pressure. This structural economy governs the rate at which new compositions can be formed and plays a central role in the system-level dynamics described in Section 8.

Boundaries are therefore not semantic objects and not independent subsystems. They are structural admissibility geometry within the population medium, required to make abstraction local, non-interpretive, dynamically regulated, and spatially negotiable.

3 The Boot Sequence: From GA to GADS

A GADS does not begin as a GADS. The transition from a standard evolutionary loop to a two-layer dynamical system occurs at a specific, identifiable moment.

3.1 The One-Layer Phase (Pre-Capture)

At initialization, the population consists of random sequences of base symbols, optionally seeded with boundary markers. No compositions exist. The system is functionally a genetic algorithm with boundary-protected regions. $\mathcal{S}(t) = \emptyset$: no compositions exist, so the service set is empty.

During this phase:

- Selection acts on raw symbol configurations.
- Boundary markers are injected probabilistically and migrate through admissible local operators in the population layer. Their positions are not fixed; delimited regions may expand, contract, or shift over time through local structural transformations.
- Boundary markers come to protect information that contributes to fitness, because organisms that do not protect useful sub-sequences lose them to mutation and crossover disruption.
- The population develops structure but has no persistent memory of that structure outside the population itself.

3.2 The Phase Transition: First Capture

When the first capture mutation fires successfully, extracting the content of a boundary-protected region and assigning it a compositional identity, the system transitions from one layer to two. A second information-carrying structure (the semantic store) now exists alongside the population.

This is the definitional boundary of a GADS: the moment the system begins acting on its own representations. At this moment $|\mathcal{S}(t)| = 1$: the captured composition remains in the organism that produced it, since capture replaces the delimited content with the new token in place.

3.3 The Two-Layer Phase (Post-Capture)

Once the first composition exists:

- Selection bias (Chain 7) immediately favors the single composition over any single base symbol for insertion and point mutations.
- The composition can appear in new organisms, reducing edit distance to configurations containing the captured sub-sequence.
- New boundary-protected regions form, potentially containing the composition.
- Further captures create higher-order compositions (compositions containing compositions).
- The full GADS dynamics engage: the composition store grows, capacity limits activate, the metabolic pipeline begins operating.

$\mathcal{S}(t)$ grows as new compositions are created and enter the population through mutation. The three-case analysis of Chain 3 has a well-defined starting condition: the system begins in Case 3 (partial cycling) and transitions toward Case 1 or Case 2 as the composition library grows relative to π_{14}.

4 Subsystem Specification

Each subsystem below is described with its theoretical role, state, operations, and PFOL anchor. The necessity of each subsystem is established by the corresponding derivation chain in Section 1.

4.1 Meta-Genome (MG) [Slow Timescale]

Derived from: Chain 2 (Persistent Semantic Store).

Theoretical Role: The persistent semantic store. It maintains the library of emergent abstractions and ensures that structural changes (deletion/inlining) do not alter the phenotypic expression.

State: A mapping $\xi : \mathcal{E} \to \Sigma^*$ where $\mathcal{E}$ is the set of encodings and Σ^* are sequences over the alphabet.

Key Operations:

- **Decode(id) $\to \sigma$:** Recursive expansion of a meta-gene identifier.
- **InlineDelete(id):** Per PFOL A3.4, replaces all references to *id* in other meta-genes with its expanded content before freeing the ID.

Top-Level-Only Usage Updates: When decoding an organism, only top-level composition tokens trigger usage signals. Each such decode request emits exactly one Usage(id) event. Nested compositions encountered during recursive expansion do not independently generate usage signals. This preserves the distinction between "used as a unit" and "happens to be embedded in a unit." A composition that only survives because it is contained inside another composition is not independently valuable. The containing composition is what is being used. Without this constraint,

inlining loses its purpose: sub-compositions that have no independent value would be kept alive artificially, consuming capacity.

This argument concerns preservation of abstraction boundaries within the MG itself and is distinct from the valuation derivation established in Chain 8.

PFOL Anchor: $\xi(e)$, $Decode(x)$, Axioms A3.2 / A3.4.

4.2 Meta-gene Creation Order (MCO) [Slow Timescale]

Derived from: Chain 7 (Recency-Biased Selection).

Theoretical Role: The chronological record of composition discovery. It defines the selection probability distribution that governs which compositions are used in mutations.

State: An ordered set $M = \{m_1, m_2, \ldots, m_k\}$.

Selection Logic: Implements an exponential decay distribution. The probability of selecting meta-gene j is

$$p(m_j) \propto \pi_{13}^{(|M|-j-1)}.$$

PFOL Anchor: Meta-gene stack M, parameter π_{13}, Equation 35.

4.3 Participation Register [Metabolic Timescale]

Derived from: Chain 3 (Finite Active Capacity), Chain 8 (Population as Energetic Engine).

Theoretical Role: Mechanical regulator of structural persistence. It maintains a bounded container of compositions whose position reflects the temporal record of participation events. It provides: (a) the capacity boundary above which the metabolic pipeline operates, and (b) the displacement ordering that feeds the deletion basket.

Mechanism

The register is a persistent ordered container of fixed capacity π_{14}. The only operation that modifies a composition's position is the binary participation signal: when a composition identifier is decoded as a top-level token during population fitness evaluation, it is moved to the retention boundary. No other signal affects position.

This constraint enforces interface-scoped participation as derived in Chain 8 and preserves the informational boundary of the register subsystem.

Selective Neutrality of Ordering

The ordering is selectively neutral. For the True-signal class (compositions decoded at least once in a generation), all members are moved to the retention boundary regardless of decode frequency. Ordering among them is neutral—none is at greater risk of displacement than any other.

For the False-signal class (compositions not decoded in a generation), ordering is the sole determinant of displacement priority. Non-participating compositions preserve their position from their last participation event. As True-signal compositions refresh, False-signal compositions drift toward the displacement boundary. The ordering does not evaluate them. It makes their non-participation legible to the displacement mechanism.

Residency vs. Service

A composition's position in the register determines only its displacement priority. It does not restrict decoding, mutation usage, capture inclusion, or any other subsystem operation. The register constrains persistence against deletion, not availability.

A composition may reside in the register without being in $\mathcal{S}(t)$—it occupies a slot but no organism currently references it. Conversely, a composition may be in $\mathcal{S}(t)$ but displaced to the deletion basket—organisms reference it, and the next decode event will generate a participation signal that rescues it back to the register.

Displacement

When a new composition enters the register and the register is at capacity, the composition at the displacement boundary (the one whose last participation signal is oldest) is moved to the deletion basket. This is not a judgment of worth. It is the mechanical consequence of finite capacity and the fact that other compositions have more recent proof of participation.

Pressure Differential

The metabolic dynamics are driven by the gap between the actual composition count and π_{14}. When the composition count is at or below π_{14}, no displacement occurs and the metabolic pipeline is idle. When the count exceeds π_{14}, each new capture forces a displacement and the deletion basket begins accumulating entries.

The relationship between $|\mathcal{S}(t)|$ and π_{14} determines which of the three dynamical cases (Chain 3) the system is operating in at any given time. This relationship fluctuates across generations.

Top-Level Usage Only

When an organism is decoded, only top-level composition references trigger participation signals. Nested components expanded during decoding do not receive independent signals. Their persistence is inherited from the top-level composition that contains them.

This argument concerns the register's informational boundary, distinct from both the valuation derivation in Chain 8 and the metabolic coherence argument in Section 4.1. If nested compositions independently generated participation signals, the register would need to disambiguate between a composition genuinely in service and one merely embedded in something in service. This would require the register to have structural knowledge of the MG's contents, violating the encapsulation between the metabolic and slow layers.

PFOL Anchor: $\mathcal{A}(t)$, $L(e,t)$, Axiom A3.3.

4.4 Deletion Basket (DB) [Metabolic Timescale]

Derived from: Chain 4 (Grace Period Under Capacity-Driven Eviction).

Theoretical Role: The selection filter. The deletion basket is where the population's binary participation signal is tested against a temporal threshold. A composition's fate in the deletion basket is determined by whether it is in the service set $\mathcal{S}(t)$: if it is, decode events will generate rescue signals; if it is not, it has no mechanism to avoid aging to the deletion threshold.

State: The Transition Set $\mathcal{T}(t)$ and age function $Age_T(e,t)$.

Logic:

- If a composition $e \in \mathcal{T}(t)$ receives a participation signal (via population decode during fitness evaluation), it is promoted back to $\mathcal{A}(t)$. Its age resets to 0. This rescue occurs because the composition is in $\mathcal{S}(t)$—organisms carrying it survived selection and were decoded.

- If $Age_T(e, t) \geq \pi_{15}$ (where $\pi_{15} = 2$), the composition has been through one complete evaluation cycle with confirmed zero participation. A DeleteRequest is sent to the MG, which performs inlining before freeing the identifier.

Minimum Threshold Derivation: Age 0 means just displaced—no information about population usage. Age 1 is ambiguous—the evaluation cycle that triggered displacement may overlap. Age 2 provides one full independent evaluation cycle with zero participation signals. This is proof of non-service, not inference. Less than 2 deletes on insufficient evidence. Greater than 2 provides redundant evidence at the cost of slower cleanup.

The DB as Selection Filter: The register provides a buffer against deletion. The deletion basket is where that buffer is removed and the composition must justify its continued existence through participation. A composition in the deletion basket that is in $\mathcal{S}(t)$ will be rescued. A composition in the deletion basket that is not in $\mathcal{S}(t)$ will be deleted. The DB does not make this determination—the population does, through the presence or absence of the participation signal. The DB merely enforces the temporal threshold that gives the signal time to arrive.

PFOL Anchor: $\mathcal{T}(t)$, Age_T, parameter $\pi_{15} = 2$.

4.5 Encoding Allocation (EA) [Metabolic Timescale]

Derived from: Chain 6 (Identifier Recycling).

Theoretical Role: Resource management and identifier conservation. Ensures the system can recycle identifiers in a finite encoding space.

State: The Reuse Pool $\mathcal{P}(t)$.

Logic: AllocateID() draws from $\mathcal{P}(t)$ if available; otherwise, it increments a global counter $H(c(t))$.

Referential Integrity: No identifier is reused while any reference to it exists. This is guaranteed by inlining (Chain 5): all references are replaced with expanded content before the identifier is freed.

PFOL Anchor: $\mathcal{P}(t)$, Axiom A2.4, Identifier Accounting A11.3.

4.6 Population (POP) [Fast Timescale]

Derived from: Chain 8 (Population as Energetic Engine).

Theoretical Role: The energetic engine. Explores the fitness landscape, discovers motifs via capture, and provides the usage signal that drives the metabolic layer.

The Fitness Gradient

A GADS requires a fitness function $F : \mathcal{X} \to \mathbb{R}$ that evaluates the phenotype. This function provides the environmental pressure that validates representational utility.

The Selection Mechanism

Differential survival generates the Usage Signal. Compositions persist only if they are carried by organisms that survive selection and are decoded during fitness evaluation. Persistence is mediated by usage during decode, not by population membership checks.

Events Emitted:

- **Capture(mapping):** Triggered by successful motif discovery.
- **Usage(id):** Triggered by top-level decode during fitness evaluation of surviving organisms.

PFOL Anchor: Predicate $InPop(x, t)$, Axioms P1–P3.

5 Multi-Timescale Coupling

The GADS architecture is defined by the interaction between three distinct temporal layers:

Layer	Subsystems	Function
Fast	Population	Stochastic search, motif discovery, usage signal generation
Metabolic	PR, DB, EA	Representation filtering, identifier recycling, forgetting
Slow	MG, MCO	Structural evolution, symbolic library, long-term memory

Dynamics of Interaction:

- **The Fast-to-Slow Ascent (Capture):** Successful fast-scale patterns are promoted to the slow scale via Capture, becoming permanent compositions.
- **The Slow-to-Fast Governance (Sampling):** The current state of the slow-scale MCO dictates the probability distribution of mutations in the fast-scale Population.
- **The Metabolic Regulator:** The Participation Register/DB subsystems act as a representational buffer, ensuring that only compositions with persistent participation across generations are maintained on the slow scale.

6 Core Interaction Pipelines

6.1 The Metabolic Pipeline (Decay and Death)

Metabolic decay is an emergent consequence of representational competition. A composition is only at risk if it falls out of the service set and is displaced by new captures.

1. **Competition:** POP generates new compositions via Capture, filling register capacity.
2. **Displacement:** When $|\mathcal{A}(t)|$ reaches π_{14} and a new composition must enter, the composition at the displacement boundary is moved to the deletion basket. Under selective neutrality of recency ordering, this will be a False-signal composition if any exist in the register.
3. **Generation Tick:** Age increments for all DB residents that do not receive a participation signal.
4. **Rescue or Death:** If the displaced composition is in $\mathcal{S}(t)$—organisms carrying it survive selection and are decoded—the participation signal fires and the composition returns to the register (age resets). If the composition is not in $\mathcal{S}(t)$, it has no mechanism to generate a rescue signal. If age reaches $\pi_{15} = 2$ with no participation signal, one full independent evaluation cycle has confirmed non-service. DeleteRequest is sent to MG.

5. **Inlining:** MG replaces all references to the composition within other compositions with its expanded content. Phenotypic identity of all referencing compositions is preserved.

6. **Recycling:** The freed identifier goes to the EA reuse pool. The composition is removed from MCO.

Partial Cycling Dynamics

When $|\mathcal{S}(t)| < \pi_{14}$, some register slots hold compositions not in the service set. If congestion is insufficient to displace them, they persist in the register but drift toward the displacement boundary as True-signal compositions refresh. This drift accumulates across generations. When congestion resumes, these compositions are displaced first, entering the deletion basket where their non-service status leads to deletion. The ordering preserves non-participation history across periods of low congestion, ensuring the metabolic pipeline has information to act on when it reactivates.

6.2 The Generative Pipeline (Discovery and Growth)

1. **Capture:** Population mutation discovers a boundary-protected region with content of at least 2 symbols. Deduplication check passes (no existing identical composition).

2. **Naming:** EA provides an identifier (from reuse pool if available, otherwise new).

3. **Definition:** MG stores the composition mapping.

4. **Registration:** MCO appends the identifier to the creation order (becomes the newest, highest-priority composition for selection).

5. **Activation:** The identifier enters the register active set.

6.3 The Closed Loop

The metabolic and generative pipelines form a complete resource cycle:

- Capture consumes boundary markers (the delimiters around the captured content are removed).
- Deletion frees identifiers that capture can reuse.
- The population generates both the usage signals that keep compositions alive and the boundary-protected regions that become new compositions.
- Compositions influence the population through selection bias (Chain 7) and decode operations.

No external resource injection is required after initialization. The system is self-sustaining.

7 Structural Locality Constraint

In a GADS, evolutionary operators and subsystem updates are strictly governed by structural locality.

Locality Principle (The Scope Constraint): A local action may only affect objects within its own system boundary, unless interaction occurs through an explicit system interface (e.g., Capture, Usage, Eviction).

Consequences of Scope:

- No operation has non-local scope; no mutation or rewrite can modify disconnected abstractions or reach across unrelated subsystems.
- There is no top-down controller or semantic oracle enforcing coherence. Coherence is emergent, not imposed.

Theoretical Role: Structural locality is the required precondition for GADS to exhibit self-regulation without an explicit controller. Without it, stability would require privileged top-down coordination. Locality ensures that the representational hierarchy is a genuine emergent product of the system's own dynamics, not a consequence of external steering.

PFOL Anchor: Delimiter Pairing Constraint (Section 2.4), Axiom A11.2.

8 Dynamics and Phase Regimes

Introduction: From Architecture to Dynamics

Sections 1–7 establish that a GADS must contain exactly six subsystems and a boundary mechanism. Those sections describe *what must exist*. This section describes *what must follow* from their interaction.

The dynamics are not additional assumptions. They are consequences of:

- Local operators acting on structured sequences (Section 3),
- Finite capacity in the Participation Register (Chain 3),
- Binary participation signals (Chain 8),
- Boundary-mediated admissibility constraints (Section 3),
- Capture consuming boundary markers (Section 7),
- Boundary-creation operations regenerating them.

The system operates across three timescales (Section 6). To isolate the dynamical contributions of each layer, we proceed incrementally. Each layer introduces one additional feedback pathway while holding others fixed.

8.1 Layer 0: Boundary Geometry Dynamics (No Capture)

Layer 0 isolates the population-level boundary mechanism while disabling capture. No compositions exist. The Participation Register, Deletion Basket, MG, EA, and MCO are inactive. The population consists solely of structured sequences containing base symbols and boundary markers.

This isolates the pure geometry of admissibility.

State Variables

Let:

- $A_0(t)$ = number of depth-0 positions (outside any boundary),
- $A_1(t)$ = number of interior positions (inside boundaries),

- $A_2(t)$ = number of boundary marker tokens.

These quantities partition the total population mass at time t.
Let:

- π_I = boundary insertion probability,
- π_R = boundary removal probability.

All other mutation operators remain local and phase-constrained but do not create persistent identity.

Structural Constraints

Layer 0 obeys the following invariants:

- Boundary markers are created and removed only in paired units.
- Boundaries do not nest.
- Mutation operators respect phase constraints (interior vs depth-0).
- No global controller enforces structure (Chain 1).

Fundamental Feedback Loop

Boundary dynamics are governed by a local insertion–removal loop:

$$\text{Insertion flux} \propto \pi_I A_0(t)$$
$$\text{Removal flux} \propto \pi_R A_2(t)$$

Insertion consumes depth-0 exposure and creates interior mass. Removal restores depth-0 exposure by eliminating boundaries.

This produces a negative feedback regulator:

- As A_2 increases, A_0 decreases.
- As A_0 decreases, insertion flux decreases.
- As A_2 increases, removal flux increases.

No valuation signal is involved. The regulator is purely geometric.

Boundary Negotiation

When boundary density increases, corridor space between adjacent boundaries decreases. Let the gap between two adjacent boundary segments be the number of depth-0 positions separating them.

As mean gap length approaches 1, boundaries begin to *negotiate* corridor space:

- Adjacent segments cannot cross.
- One segment may shrink while the other expands.

- Local operators preserve boundary pairing and non-interpenetration constraints.

Negotiation is activated when:

$$\text{Mean gap length} \to 1$$

This produces a crowding regime in which boundaries redistribute local space without global merging.

Observed Phase Regimes

Simulation of Layer 0 under varying (π_I, π_R) reveals four distinct regimes:

(A) Balanced Regime When insertion and removal fluxes are comparable:

- A_2 stabilizes at intermediate density,
- A_1 stabilizes proportionally,
- A_0 remains non-zero,
- negotiation events are rare.

This is a steady structural equilibrium.

(B) Negotiation Regime When insertion dominates moderately:

- A_2 increases,
- corridor length decreases,
- fraction of gaps ≤ 1 increases,
- segment lengths fluctuate locally but mean segment length remains bounded.

Boundaries redistribute space rather than merge. Global takeover does not occur.

(C) Saturation Regime When insertion strongly dominates:

- A_2 overshoots,
- A_0 collapses,
- negotiation becomes persistent,
- removal flux eventually spikes due to high A_2.

This produces a structural boom–bust cycle: expansion followed by collapse.

(D) Removal-Dominant Regime When removal dominates:

- $A_2 \to 0$,
- $A_1 \to 0$,
- the system reverts to a boundary-free medium.

No higher structure persists.

Layer 0 Conclusion

Layer 0 demonstrates that:

- A self-regulating structural economy emerges from purely local boundary operators.
- Negative feedback exists without valuation or memory.
- Crowding produces redistribution (negotiation) rather than collapse.
- Overshoot can generate structural boom–bust cycles.

These dynamics arise before identity, before participation, and before the metabolic pipeline activates.

Layer 0 therefore establishes the geometric substrate upon which capture and representational evolution operate.

Subsequent layers will introduce identity, capacity, and participation while preserving this underlying structural regulator.

9 PFOL Alignment Summary

Every subsystem and mechanism specified in this document corresponds to a formally verified element of the MEGA PFOL specification:

GADS Component	**Derivation**	**PFOL Reference**
Structural Locality	Chain 1	Section 2.4, A11.2
Meta-Genome (MG)	Chain 2	$\xi(e)$, A3.2, A3.4
Participation Register	Chain 3	$\mathcal{A}(t)$, A3.3
Deletion Basket (DB)	Chain 4	$\mathcal{T}(t)$, Age_T, π_{15}
Inlining	Chain 5	A3.4
Encoding Allocation (EA)	Chain 6	$\mathcal{P}(t)$, A2.4, A11.3
Creation Order (MCO)	Chain 7	M, π_{13}, Eq. 35
Population (POP)	Chain 8	$InPop(x,t)$, P1–P3
Boundary Mechanism	Section 3	Section 2.4, P2